

Torsion balances as operational probes of semiclassical gravity: Matched-filter bounds, torque-diffusion constraints, and quantum-noise benchmarks

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Calibrated torsion-balance spectra can be mapped to bounds on deterministic and stochastic departures from standard gravitational sourcing rules in the weak-field Newtonian regime. Using one-sided spectra, we formulate a linear calibrated angle-equivalent output-noise budget and the finite-time matched-filter/Cramér–Rao bound for a known torque template. For stochastic models, the calibrated standard noise budget—thermal, Newtonian, environmental, imprecision, and backaction terms—is subtracted from the observed angle spectrum to define a residual spectrum, which is then converted to an equivalent residual torque spectrum using the calibrated torsional susceptibility. This residual provides an upper bound on any additional stationary stochastic torque noise. Only in the Markovian white-noise limit can this frequency-resolved bound be compressed to a single torque-diffusion coefficient D_τ ; colored or non-Markovian models must be compared with the full residual spectrum. Page–Geilker branch discrimination and Fedida–Kent mixture-equivalence tests address different physical sourcing questions, but after their competing predictions are projected onto torsion-balance torque templates they become the same statistical problem of matched-filter template discrimination. For a representative room-temperature Cavendish benchmark, the resonant thermal angle ASD is 1.36×10^{-4} rad/ $\sqrt{\text{Hz}}$, whereas the measurement-added SQL is 3.25×10^{-12} rad/ $\sqrt{\text{Hz}}$. For the torsion-balance search of Yan *et al.*, the reported $0.3 \mu\text{rad}/\sqrt{\text{Hz}}$ sensitivity at 2.5 mHz implies a conservative observed-floor bound $D_\tau \lesssim 2.4 \times 10^{-23} \text{ N}^2 \text{ m}^2 \text{ s}$ when that public floor is interpreted as white torque noise. These expressions provide a common interface between calibrated torsion-balance data, deterministic template tests, and stochastic semiclassical-gravity searches without claiming a direct test of the full relativistic semiclassical Einstein equation.

I. INTRODUCTION

Torsion balances remain laboratory instruments for measurements of small gravitational torques. They have long been used in determinations of Newton’s constant [1–3]. They also provide null tests of the weak equivalence principle and of short-range departures from the inverse-square law [4–6]. The Page–Geilker experiment used a torsion balance to confront branch-dependent semiclassical sourcing in a macroscopic setting [7]. More recent proposals and experiments use torsional or related macroscopic mechanical systems as interfaces between quantum measurement and weak gravity [8, 9].

A torsion-balance analysis of semiclassical gravity must keep three literatures conceptually separate. Linear quantum measurement theory fixes the imprecision–backaction tradeoff and the measurement-added standard quantum limit (SQL) [10–14]. Torsion-balance metrology fixes the calibrated mechanical response and the ordinary thermal, anelastic, seismic, Newtonian-noise, and technical backgrounds; a separate preprint analyzes atmospheric Newtonian noise in torsion-balance determinations of G within a GUM-style framework [15]. Semiclassical and postquantum-gravity models then en-

ter only after that measurement interface has been specified, through deterministic branch-dependent torques, state-preparation-dependent sourcing rules, or additional stochastic torque noise [7, 16–19].

The present paper focuses on this last interface. Big- G metrology, WEP and inverse-square-law null tests, and quantum-noise budgeting are not treated as new physical results here; they provide the calibrated language needed to translate an observed angle spectrum into bounds on torque templates and residual stochastic torque spectra.

Throughout the paper, a *sourcing rule* means a physical or theoretical map from a source preparation—for example a mass-density history, a quantum state, a branch label, or a reduced density operator—to the weak gravitational field and hence to the torque template acting on the balance. It is not a synonym for an experimental error source. Calibration uncertainties, GUM-style uncertainty budgets, Newtonian-noise backgrounds, and other technical disturbances enter separately through the measured spectrum and the residual construction below.

The paper makes four modest contributions. First, it standardizes the measurement interface by keeping one-sided PSD conventions, transfer-function factors, and angle-to-torque conversion explicit from the output spectrum to the final constraint. These are standard tools, but their consistent use prevents ambiguities when different gravitational sourcing models are

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compared with torsion-balance data. Second, it applies the standard matched-filter/Cramér–Rao formalism to finite-time torsion-balance torque templates for semiclassical and postquantum-gravity tests. The data-analysis method is not new; the contribution is the explicit projection of candidate sourcing rules onto calibrated torque templates. Third, it separates a general residual equivalent-torque spectrum from the special Markovian white-noise compression into a single experimental summary number D_τ . For colored or non-Markovian models, the spectrum rather than D_τ is the comparison object. Fourth, Page–Geilker branch discrimination and the mixture-equivalence-principle (MEP) discussion of Fedida and Kent are presented as interpretive applications of the same template formalism. The two tests address different physical sourcing questions and are not physically equivalent; their commonality appears only after the competing predictions are projected onto torsion-balance torque templates.

Related collapse-model and Schrödinger–Newton studies also require geometry-dependent force or torque spectra before comparison with data [20–25]. The present paper addresses the semiclassical/postquantum-gravity side of the same experimental interface [17–19].

The paper is organized as follows. Section II summarizes the torsion-balance response, fixes the one-sided conventions, and states the linear calibrated output spectrum. Section III treats deterministic torque templates and the finite-time matched-filter bound. Section IV translates residual spectra into stochastic torque-noise bounds and, only in the white limit, torque-diffusion constraints. Section V provides quantitative scale-setting benchmarks for a representative Cavendish platform and for the recent torsion-balance search of Yan *et al.* [8]. Section VI treats Page–Geilker branch tests and proper/improper-mixture MEP tests as interpretive applications of the torque-template formalism. Section VII collects experimental design implications. A generic two-channel variational-readout strategy is summarized in Appendix D; it is included as an extension of the noise model, not as the central result.

II. TORSION-BALANCE RESPONSE AND SPECTRAL CONVENTIONS

A. Mechanical response and notation

For small angular displacement θ , a torsion balance is modeled as a linear torsional oscillator with moment of inertia I , torsional spring constant κ , and damping rate γ :

$$I\ddot{\theta}(t) + I\gamma\dot{\theta}(t) + \kappa\theta(t) = \tau_x(t) + \tau_{\text{th}}(t) + \tau_{\text{NN}}(t) + \tau_{\text{env}}(t) + \tau_{\text{ba}}(t). \quad (1)$$

Here $\tau_x(t)$ denotes whichever signal torque one wishes to constrain: the usual Newtonian torque in a Big- G measurement, a Yukawa or WEP-violating template,

or a branch-dependent or stochastic semiclassical contribution. The term τ_{NN} denotes Newtonian noise, i.e. gravity-gradient torque from fluctuating environmental mass distributions. It may be grouped inside a broader environmental torque τ_{env} , but we keep it explicit when discussing low-frequency backgrounds and auxiliary-channel subtraction. With the Fourier convention $f(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} f(\omega) e^{-i\omega t}$, the torsional susceptibility is

$$\chi(\omega) = \frac{1}{\kappa - I\omega^2 - iI\gamma\omega}, \quad (2)$$

so that $\theta(\omega) = \chi(\omega)\tau(\omega)$ and $\omega_0 = \sqrt{\kappa/I}$, $Q = \omega_0/\gamma$.

We use one-sided power spectral densities (PSDs) for $\omega > 0$ throughout. The corresponding amplitude spectral density (ASD) is $\sqrt{S(\omega)}$. Table I summarizes the quantities used most often in the manuscript. Angles are quoted in radians for readability, although rad is dimensionless in SI.

For any real stationary process $x(t)$ we use the one-sided/symmetrized relation $S_{xx}^{1s}(\omega) = 2S_{xx}^{\text{sym}}(\omega)$ for $\omega > 0$. In particular, $S_{\theta\theta}^{\text{SQL},1s} = 2\hbar|\chi|$ corresponds to $S_{\theta\theta}^{\text{SQL,sym}} = \hbar|\chi|$. We reserve $\chi(\omega)$ for the calibrated complex susceptibility of the torsional mode under study and $|\chi(\omega)|$ for its magnitude. When a specific structural-loss model is invoked we write $\chi_\phi(\omega)$ explicitly.

For later quantum benchmarks we also introduce the torsional Hamiltonian

$$\hat{H} = \frac{\hat{L}^2}{2I} + \frac{\kappa}{2}\hat{\theta}^2, \quad [\hat{\theta}, \hat{L}] = i\hbar, \quad (3)$$

with zero-point angle

$$\theta_{\text{zpf}} = \sqrt{\frac{\hbar}{2I\omega_0}}. \quad (4)$$

B. Operational weak-field sourcing and torque projection

The word “semiclassical” is used here in an operational weak-field sense. The relativistic semiclassical Einstein equation is the motivation, but a laboratory torsion balance in the slow-motion, weak-field regime is described by the Newtonian projection of a candidate sourcing rule. For a rule x we write schematically

$$\nabla^2\Phi_x(\mathbf{r}, t) = 4\pi G\rho_x^{\text{eff}}(\mathbf{r}, t), \quad (5)$$

where ρ_x^{eff} may be an ordinary classical density, an expectation value, a branch-conditioned density, or another model-dependent effective source. The balance does not measure Φ_x directly. It measures the torque obtained by projecting the weak field onto the probe mass distribution,

$$\tau_x(t) = \int d^3r \rho_{\text{probe}}(\mathbf{r}) [\mathbf{r} \times (-\nabla\Phi_x(\mathbf{r}, t))]_z, \quad (6)$$

TABLE I. Symbols and one-sided conventions used in the main text.

Symbol	Meaning	Units
$\chi(\omega)$	calibrated torsional susceptibility	rad/(N m)
$S_{\theta\theta}(\omega)$	angle PSD	rad ² /Hz
$S_{\tau\tau}(\omega)$	torque PSD	N ² m ² /Hz
$\sqrt{S_{\theta\theta}}$	angle ASD	rad/ $\sqrt{\text{Hz}}$
$\sqrt{S_{\tau\tau}}$	torque ASD	N m/ $\sqrt{\text{Hz}}$
D_τ	stochastic-process white torque-diffusion coefficient, defined by $S_{\tau\tau} = 4D_\tau$	N ² m ² s
B_w	effective noise bandwidth of the analysis window	Hz

and the observed angle is the calibrated response $\theta_x(\omega) = \chi(\omega)\tau_x(\omega)$. This is the sense in which torsion-balance data probe semiclassical or postquantum sourcing rules in this paper: they constrain their observable torque projections, not the full relativistic field equation itself.

We also distinguish environmental Newtonian noise from a formal fluctuation of Newton's constant. A replacement $G \rightarrow G + \delta G$ can be useful as a theoretical scale-setting exercise in Schrödinger–Newton-type models, but it is not the model of Newtonian noise used below. Environmental Newtonian noise is generated by fluctuating mass distributions and appears as an additive torque, a correlated auxiliary-channel contribution, or a multiplicative gravity-gradient perturbation. If the torsional degree of freedom is quantized, any such scale-setting comparison should be made with the angular Hamiltonian in Eq. (3), rather than with a free center-of-mass Hamiltonian $\hat{P}^2/(2M)$.

C. Linear calibrated output spectrum and cross correlations

We write the measured, angle-equivalent readout as

$$y(\omega) = \theta(\omega) + n_\theta(\omega), \quad (7)$$

where $n_\theta \equiv \theta_{\text{imp}}$ is the angle-referred imprecision noise. Let $a, b \in \{\text{th}, \text{NN}, \text{env}, \text{ba}\}$ label thermal torque, Newtonian-noise torque, other environmental torque, and measurement backaction torque. The most general stationary one-sided output PSD at the linear level is

$$S_{yy}(\omega) = S_{n_\theta n_\theta}(\omega) + |\chi(\omega)|^2 \sum_{a,b} S_{\tau_a \tau_b}(\omega) + 2 \text{Re} \left[\chi(\omega) \sum_a S_{n_\theta \tau_a}(\omega) \right]. \quad (8)$$

This equation is the bookkeeping answer to where Newtonian noise enters: as an additive gravity-gradient torque τ_{NN} , and, if it is correlated with some other channel, through the corresponding cross spectrum. For example, a direct correlation between a gravity-gradient background and the angle readout would appear as

$2\text{Re}[\chi S_{n_\theta \tau_{\text{NN}}}]$, while correlations between two torque channels appear inside the double sum.

For the main formulas we use the standard diagonal approximation: thermal noise, Newtonian noise, other environmental torque noise, and optical backaction are treated as mutually independent, except for the deliberately engineered or measured imprecision–backaction correlation. Then Eq. (8) reduces to

$$S_{yy}(\omega) = S_{\theta\theta}^{\text{imp}}(\omega) + |\chi(\omega)|^2 \left[S_{\tau\tau}^{\text{th}}(\omega) + S_{\tau\tau}^{\text{NN}}(\omega) \right] + |\chi(\omega)|^2 \left[S_{\tau\tau}^{\text{env}}(\omega) + S_{\tau\tau}^{\text{ba}}(\omega) \right] + 2 \text{Re}[\chi(\omega) S_{n_\theta \tau_{\text{ba}}}(\omega)]. \quad (9)$$

Equation (9) is the angle-equivalent noise budget used in the rest of the paper: *every* additive torque-noise PSD enters the measured angle through the same transfer function $|\chi|^2$. This is elementary from $\theta = \chi\tau$, but it is the key bookkeeping identity. When correlations other than $S_{n_\theta \tau_{\text{ba}}}$ are non-negligible, one should return to Eq. (8) and calibrate the relevant cross spectra. Appendix D records the corresponding generic two-channel estimator for the case in which correlations are exploited rather than merely subtracted.

For a quantum-limited linear detector the one-sided spectra obey

$$S_{\theta\theta}^{\text{imp}}(\omega) S_{\tau\tau}^{\text{ba}}(\omega) - |S_{n_\theta \tau_{\text{ba}}}(\omega)|^2 \geq \hbar^2. \quad (10)$$

In the uncorrelated case $S_{n_\theta \tau_{\text{ba}}} = 0$, minimizing the measurement-added part

$$S_{\theta\theta}^{\text{add}}(\omega) = S_{\theta\theta}^{\text{imp}}(\omega) + |\chi(\omega)|^2 S_{\tau\tau}^{\text{ba}}(\omega) \quad (11)$$

gives the standard quantum limit

$$S_{\theta\theta}^{\text{SQL}}(\omega) = 2\hbar|\chi(\omega)|. \quad (12)$$

Because later sections compare thermal, classical, and measurement-added terms on the same one-sided plots, we explicitly retain the one-sided factors here rather than leaving them implicit.

D. Thermal torque noise: viscous and structural damping

For viscous damping, the fluctuation–dissipation theorem gives a white thermal torque PSD,

$$S_{\tau\tau}^{\text{th,visc}}(\omega) = 4k_B T I \gamma, \quad (13)$$

and therefore $S_{\theta\theta}^{\text{th,visc}} = |\chi|^2 S_{\tau\tau}^{\text{th,visc}}$.

Real torsion fibers are often better described by a structural-loss model with a frequency-independent loss angle ϕ . For the Fourier convention above we write $\kappa \rightarrow \kappa(1 - i\phi)$ and define

$$\chi_\phi(\omega) = \frac{1}{\kappa(1 - i\phi) - I\omega^2}. \quad (14)$$

The one-sided angle thermal PSD is then

$$S_{\theta\theta}^{\text{th,str}}(\omega) = \frac{4k_B T}{\omega} \text{Im} \chi_\phi(\omega). \quad (15)$$

This form has the correct dimensions and shows that the effective drive noise is no longer white. Indeed, the angle spectrum can be re-expressed as an equivalent torque PSD

$$S_{\tau\tau}^{\text{eq,str}}(\omega) \equiv \frac{S_{\theta\theta}^{\text{th,str}}(\omega)}{|\chi_\phi(\omega)|^2} = \frac{4k_B T \kappa \phi}{\omega}, \quad (16)$$

which exhibits the $1/\omega$ scaling directly. Within the constant- ϕ response model this conversion is exact algebraically; the caution concerns the validity range of the model itself, not the equivalent-torque conversion. This distinction matters below, because a stochastic semiclassical torque source is compared to the equivalent torque ASD rather than to the raw angle ASD. At the same time, Eqs. (15) and (16) should be read as an effective band-limited parameterization: a strictly frequency-independent loss angle extrapolated to arbitrarily low frequency produces the familiar divergence of the equivalent drive noise and is not a fully causal description in the $\omega \rightarrow 0$ limit. In practice, one should use the constant- ϕ model only over the band where it is independently validated for the suspension in question [26].

E. Additive Newtonian noise, multiplicative gradients, and resonance drift

The damping rate γ in Eq. (1) is an effective mechanical loss parameter: it may include gas damping, internal fiber loss, anelastic loss, or feedback damping. It should not be interpreted as a “gravitational viscosity.” A static external gravity gradient instead changes the conservative torsional stiffness, for example $\kappa \rightarrow \kappa + \kappa_g$, and hence shifts the resonance frequency. Fluctuating environmental mass distributions can therefore enter in two distinct ways. The first is the additive Newtonian-noise

torque $\tau_{\text{NN}}^{\text{add}}(t)$ already included in Eq. (1). The second is a multiplicative stiffness fluctuation,

$$I\ddot{\theta} + I\gamma\dot{\theta} + [\kappa + \delta\kappa_{\text{NN}}(t)]\theta = \tau_x + \tau_{\text{NN}}^{\text{add}} + \dots \quad (17)$$

The factor of one half in the resonance drift follows directly from expanding the instantaneous frequency,

$$\begin{aligned} \omega_0(t) &= \sqrt{\frac{\kappa + \delta\kappa_{\text{NN}}(t)}{I}} = \omega_0 \sqrt{1 + \frac{\delta\kappa_{\text{NN}}(t)}{\kappa}} \\ &\simeq \omega_0 \left[1 + \frac{1}{2} \frac{\delta\kappa_{\text{NN}}(t)}{\kappa} - \frac{1}{8} \left(\frac{\delta\kappa_{\text{NN}}(t)}{\kappa} \right)^2 + \dots \right], \end{aligned} \quad (18)$$

so that, for $|\delta\kappa_{\text{NN}}| \ll \kappa$,

$$\frac{\delta\omega_0(t)}{\omega_0} \simeq \frac{1}{2} \frac{\delta\kappa_{\text{NN}}(t)}{\kappa}. \quad (19)$$

For example, a fractional stiffness perturbation $|\delta\kappa_{\text{NN}}|/\kappa = 10^{-4}$ produces a fractional resonance shift $|\delta\omega_0|/\omega_0 \simeq 5 \times 10^{-5}$. The perturbative analysis assumes both $|\delta\kappa_{\text{NN}}|/\kappa \ll 1$ and stability, $\kappa + \delta\kappa_{\text{NN}} > 0$. If a transient gradient pushed the system close to $\delta\kappa_{\text{NN}} = -\kappa$, the linear susceptibility used in the matched filter would no longer be the correct calibrated response; such intervals should be modeled separately or vetoed, not interpreted as an unbounded Newtonian-noise contribution.

The same stiffness perturbation also shifts the static angular operating point. For a static calibration torque τ_{static} , the instantaneous equilibrium is

$$\begin{aligned} \theta_{\text{eq}}(t) &= \frac{\tau_{\text{static}}}{\kappa + \delta\kappa_{\text{NN}}(t)} \simeq \theta_0 \left[1 - \frac{\delta\kappa_{\text{NN}}(t)}{\kappa} \right], \\ \theta_0 &\equiv \frac{\tau_{\text{static}}}{\kappa}. \end{aligned} \quad (20)$$

Thus $\Delta\theta_{\text{eq}}/\theta_0 \simeq -\delta\kappa_{\text{NN}}/\kappa$: a positive gravity-gradient stiffness reduces the deflection, while a negative stiffness perturbation increases it. Unlike an additive torque PSD, this multiplicative term convolves the gradient fluctuation with the oscillator coordinate in frequency space. In a narrowband analysis it is usually handled as a calibration uncertainty in $\chi(\omega)$, a slow line-frequency drift, or a reduction of the coherent matched-filter response. Equivalently, it can be monitored with gravity-gradient, barometric, hydrological, or tilt channels and propagated into the residual through the same multichannel calibration used for additive Newtonian noise. The formulas below assume that the susceptibility entering the analysis is the calibrated response over the relevant integration interval; if the resonance drifts appreciably, the quoted S_{yy} and the leakage factor in Eq. (28) must include that drift.

III. DETERMINISTIC TORQUE TEMPLATES AND FINITE-TIME MATCHED FILTERING

A. General template amplitude

We parameterize a deterministic signal of interest as

$$\tau_x(t) = g_x \Lambda_x(t), \quad (21)$$

where $\Lambda_x(t)$ is known and g_x is the unknown amplitude parameter to be constrained. The usual Newtonian torque in a Big- G measurement is obtained by setting $g_x = G$. A Yukawa deviation at fixed range λ may be written as $g_x = \alpha$ with the torque template $\Lambda_{\alpha,\lambda}(t)$ determined by the attractor geometry. A Page–Geilker branch comparison corresponds to $g_x = \pm 1$ and $\Lambda_x(t) = \tau_0(t)$. The advantage of Eq. (21) is that all of these cases share the same estimator. The “incoming gravity signal” in this language is the calibrated torque waveform produced by the chosen source preparation and modulation protocol.

For a rotating or periodically translated attractor, the template is usually a discrete Fourier series,

$$\Lambda_x(t) = \sum_{k \geq 1} \Lambda_k \cos(k\Omega_{\text{rot}}t + \varphi_k), \quad (22)$$

where Ω_{rot} is the mechanical modulation frequency and the nonzero harmonics are fixed by the source geometry. If the attractor has an N -fold symmetry, the first useful line is often at $\omega_m = N\Omega_{\text{rot}}$, with additional harmonics depending on the mass pattern and shielding. Thus there is no universal gravitational-signal frequency; the operating frequencies are the Fourier lines of the source torque template that are intentionally placed inside the calibrated torsion-balance band.

The angle response to the template is

$$\theta_x(\omega) = g_x \chi(\omega) \Lambda_x(\omega). \quad (23)$$

For stationary Gaussian noise with one-sided PSD S_{yy} , the Fisher information for g_x is

$$\mathcal{I}(g_x) = 4 \int_0^\infty \frac{d\omega}{2\pi} \frac{|\chi(\omega) \Lambda_x(\omega)|^2}{S_{yy}(\omega)}, \quad (24)$$

and the Cramér–Rao bound is

$$\delta g_x \geq \mathcal{I}(g_x)^{-1/2}. \quad (25)$$

Equation (24) is the central deterministic result of the paper. It maps any calibrated output spectrum to a bound on the amplitude of any known torque waveform.

B. Single-tone modulation and explicit window factors

Suppose

$$\Lambda_x(t) = \Lambda_0 \cos(\omega_m t + \varphi) \quad (26)$$

is observed for a finite integration time T_{int} . If the analysis is performed with a window $w(t)$, the relevant order-one factor is the equivalent noise bandwidth

$$B_w = \frac{\int_0^{T_{\text{int}}} dt w^2(t)}{\left[\int_0^{T_{\text{int}}} dt w(t) \right]^2} = \frac{\eta_w}{T_{\text{int}}}, \quad (27)$$

with $\eta_{\text{rect}} = 1$ for a rectangular window and $\eta_{\text{Hann}} = 3/2$ for a Hann window. When the modulation frequency is tracked well enough that leakage is negligible, the amplitude bound becomes

$$\frac{\delta g_x}{|g_x|} \simeq \frac{\sqrt{S_{yy}(\omega_m) B_w}}{|g_x| |\chi(\omega_m)| |\Lambda_0|}. \quad (28)$$

The approximate $T_{\text{int}}^{-1/2}$ scaling follows from coherent statistical averaging of a stationary narrowband process. It should not be interpreted as a thermodynamic reduction of the entropy of the local oscillator or of the gravitational environment. It also holds only while the noise floor and the response remain approximately stationary over the analysis window. If slow Newtonian-noise backgrounds, tilt, thermal drift, or multiplicative stiffness fluctuations change the mean, the correlation structure, or the resonance frequency during the run, increasing T_{int} need not improve the uncertainty with the ideal square-root law.

If the phase is also unknown and must be estimated from two orthogonal quadratures, the right-hand side acquires the usual factor $\sqrt{2}$. If slow eigenfrequency drift or template mismatch reduces the coherent response, one should further replace $|\Lambda_0| \rightarrow \mathcal{L}|\Lambda_0|$ with a leakage factor $\mathcal{L} \leq 1$. A useful definition is the normalized overlap between the windowed true signal $s(t)$ and the windowed template $h(t)$,

$$\mathcal{L} = \frac{\left| \int_0^{T_{\text{int}}} dt w(t) s(t) h(t) \right|}{\left[\int_0^{T_{\text{int}}} dt w^2(t) s^2(t) \int_0^{T_{\text{int}}} dt w^2(t) h^2(t) \right]^{1/2}}. \quad (29)$$

For a rectangular window and a pure frequency offset $\Delta\omega$ this reduces, in the many-cycle approximation, to $\mathcal{L} \simeq |\text{sinc}(\Delta\omega T_{\text{int}}/2)|$; for a Hann or other window it is the corresponding normalized window-transform overlap. In real millihertz analyses, nonstationary drifts can also raise the local floor $S_{yy}(\omega_m)$ itself through spectral leakage into the analysis band. The conservative procedure is therefore to estimate $S_{yy}(\omega_m)$ from data processed with the same windowing, segmentation, line-removal, and drift-tracking steps used in the matched filter.

Equation (28) is the precise finite-time version of the qualitative statement that one should maximize $|\chi|^2/S_{yy}$ within the trusted analysis band. If the dominant background is white torque noise, the best band is near resonance. If the dominant background is drift or structural-loss torque noise, the optimum shifts upward. In all cases the relevant quantity is the measured spectrum itself rather than a generic appeal to “working near resonance.”

C. Operational band and noise-frequency taxonomy

The analysis is not intrinsically restricted to the ultralow-frequency band. The useful operating band for a given experiment is the set of frequencies where (i) the signal template has support, (ii) the susceptibility and auxiliary-channel calibrations are reliable, and (iii) the Fisher kernel

$$\mathcal{K}_x(\omega) \equiv \frac{|\chi(\omega)\Lambda_x(\omega)|^2}{S_{yy}(\omega)} \quad (30)$$

is large. Millihertz operation is natural for many torsion balances because the fundamental torsional resonances are often in that range and slow source modulation is technically clean, but the same equations apply to higher-frequency modulation until the torsional response, internal modes, or readout noise make $\mathcal{K}_x$ small.

Representative examples of the standard terms are cited at the points where they enter the model: torsion-balance metrology and structural thermal noise in Refs. [1, 2, 26], linear quantum measurement and optomechanical readout in Refs. [8, 10–12], and environmental Newtonian noise in Refs. [15, 27]. Table II therefore does not try to assign universal numerical frequency bands. Such bands are apparatus and site dependent. Instead, it records the diagnostic role of each component in the calibrated one-sided spectrum.

Thus the practical design question is to identify an operational window, not to assume in advance that the lowest accessible frequencies are always best. If Newtonian noise dominates a millihertz band, moving the template to a higher harmonic can help; if the response has already rolled off and imprecision dominates, moving still higher can hurt. This tradeoff is the torsion-balance analogue of the noise-budget optimization used in precision interferometers: the classical environmental, thermal, and quantum measurement terms must be separated before the best science band is obvious.

D. Deterministic null tests: WEP and inverse-square-law searches

The same template language immediately covers the standard torsion-balance null tests. In a WEP experiment the signal of interest is a composition-dependent differential torque; in an inverse-square-law search the signal is the Yukawa template generated by

$$V(r) = -\frac{Gm_1m_2}{r} \left[1 + \alpha e^{-r/\lambda} \right]. \quad (31)$$

The experimental result is therefore a bound on g_x with the appropriate Λ_x . Our point is not that WEP or Yukawa searches are new, but that they furnish classical-source calibration data against which any semiclassical or postquantum model must reduce to the standard macroscopic limit.

IV. STOCHASTIC SEMICLASSICAL GRAVITY AS RESIDUAL TORQUE NOISE

A. Residual spectra and model-independent bounds

Consistent classical-quantum gravity frameworks generically predict that the classical sector must be noisy or diffusive when coupled to quantum matter [17, 18]. At torsion-balance level the most economical phenomenology is therefore an additional stationary torque-noise term with one-sided PSD $S_{\tau\tau}^{\text{sc}}(\omega)$:

$$S_{yy}^{\text{obs}}(\omega) = S_{yy}^{\text{std}}(\omega) + |\chi(\omega)|^2 S_{\tau\tau}^{\text{sc}}(\omega), \quad (32)$$

where S_{yy}^{std} contains the calibrated standard contributions: thermal suspension noise, Newtonian and other environmental torque backgrounds, readout imprecision, backaction, calibration uncertainty, and any validated cross spectra. The candidate semiclassical or postquantum stochastic term is not identified with these ordinary stationary backgrounds; it is an additional residual term after the standard budget has been accounted for.

Formally, the residual estimate is

$$\hat{S}_{yy}^{\text{res}}(\omega) \equiv \hat{S}_{yy}^{\text{obs}}(\omega) - \hat{S}_{yy}^{\text{std}}(\omega), \quad (33)$$

and a physical excess-noise interpretation gives the pointwise upper bound

$$S_{\tau\tau}^{\text{sc}}(\omega) \leq \frac{S_{yy}^{\text{res},+}(\omega)}{|\chi(\omega)|^2},$$

$$S_{yy}^{\text{res},+} \equiv \max\{\hat{S}_{yy}^{\text{res}}, 0\} \quad \text{or an upper confidence band.} \quad (34)$$

Equation (33) is simple algebraically but difficult experimentally. The standard model must include thermal noise, Newtonian noise, environmental drift, seismic tilt, readout imprecision, backaction, calibration uncertainty, and other technical disturbances. If any component is incomplete or over-subtracted, individual frequency bins of $\hat{S}_{yy}^{\text{res}}$ can become negative or falsely positive. In practice one should quote uncertainty bands, confidence intervals, or band-averaged limits rather than treating the residual as an exact positive function at every frequency. When a fully subtracted residual is unavailable, Eq. (34) can still be used conservatively by replacing the residual upper band with the published total observed floor.

For colored low-frequency searches the main challenge is often the construction of the residual itself: atmospheric pressure, hydrological motion, seismic displacement, tilt, or local moving masses can generate broad $1/f^n$, Lorentzian, or otherwise colored Newtonian-noise spectra that mimic non-Markovian semiclassical torque noise. Without auxiliary environmental channels and cross-spectral subtraction, such a residual should be interpreted as an upper bound on additional stochastic torque noise, not as evidence for a new gravitational-noise source.

TABLE II. Diagnostic map of the main one-sided spectral components in a calibrated torsion-balance analysis. The table is not a universal frequency taxonomy; the relevant band is determined by the measured local floor S_{yy} , the calibrated susceptibility χ , and the torque template Λ_x .

Component	Angle-equivalent entry	Common spectral character	Interpretation here
Thermal suspension noise	$ \chi ^2 S_{\tau\tau}^{\text{th}}$; Eq. (13) for viscous loss and Eq. (16) for structural loss.	White torque noise for viscous damping; $1/\omega$ equivalent torque noise for constant structural loss.	Calibrated standard floor. It must be included in S_{yy}^{std} , not counted as an exotic residual.
Newtonian/environmental gravity gradients	$ \chi ^2 S_{\tau\tau}^{\text{NN}}$ for additive torque noise; $\delta\kappa_{\text{NN}}(t)$ for multiplicative stiffness drift.	Usually colored at low frequency and site dependent; may contain deterministic transients or coherent lines.	Standard environmental background unless deliberately used as a deterministic torque template or auxiliary-channel subtraction target.
Readout imprecision	$S_{\theta\theta}^{\text{imp}}$ in angle units; $S_{\theta\theta}^{\text{imp}}/ \chi ^2$ in equivalent torque units.	Often approximately white over a local readout band; torque-equivalent impact grows where $ \chi $ is small.	Part of the measured local floor and the measurement-added noise budget.
Backaction and calibrated correlations	$ \chi ^2 S_{\tau\tau}^{\text{ba}} + 2\text{Re}[\chi S_{n\theta\tau\text{ba}}]$, with possible auxiliary cross spectra.	Detector dependent; constrained by the imprecision–backaction inequality and altered by variational readout.	Measurement-added noise. It can be optimized or subtracted only when the relevant cross spectra are calibrated.
Deterministic gravity template	$g_x \chi(\omega) \Lambda_x(\omega)$ in the measured angle channel.	Known waveform, rotating-source harmonics, or narrow coherent lines.	Estimated with the matched-filter/Cramér–Rao statistic; covers Big- G , WEP/ISL, Page–Geilker, and MEP templates after projection.
Candidate stochastic gravitational torque	$ \chi ^2 S_{\tau\tau}^{\text{sc}}(\omega)$ added only after the standard budget is specified.	Model dependent: white, Lorentzian, $1/f^n$, OU-type, or other non-Markovian spectra.	Constrained as a residual equivalent-torque spectrum; compressible to D_τ only in the Markovian white-noise limit.

B. White torque diffusion

For a white stochastic torque process we define D_τ by the one-sided relation

$$S_{\tau\tau}^{\text{sc}}(\omega) = 4D_\tau. \quad (35)$$

Here “diffusion coefficient” is used in the stochastic-process sense: a white random torque causes diffusive growth of the angular-momentum variance, as shown in Appendix B. It is not a spatial diffusion constant and does not mean that a field diffuses through space or decays in amplitude. This scalar compression applies only in the Markovian white-noise limit. If a candidate semiclassical, postquantum, or collapse-inspired model predicts colored or non-Markovian torque noise—for example $1/f$, $1/f^2$, Lorentzian, or Ornstein–Uhlenbeck-type spectra—the correct comparison is not to a single D_τ but directly to the frequency-resolved bound in Eq. (34) or its band-averaged version in Eq. (37). Equations (34) and (35) then imply

$$D_\tau \leq \frac{S_{yy}^{\text{res},+}(\omega)}{4|\chi(\omega)|^2} \quad (\text{pointwise bound}). \quad (36)$$

A band-averaged version is obtained by weighting each frequency bin by the estimator used in the actual analysis:

$$D_\tau \leq \frac{\int_{\mathcal{B}} \frac{d\omega}{2\pi} W(\omega) S_{yy}^{\text{res},+}(\omega)/|\chi(\omega)|^2}{4 \int_{\mathcal{B}} \frac{d\omega}{2\pi} W(\omega)}, \quad (37)$$

with $W(\omega) \geq 0$ chosen, for instance, to emphasize the band where the residual spectrum is best controlled. Equations (36) and (37) are intentionally model-independent at the spectrum level. They make no assumptions about the microscopic origin of the stochastic

term beyond stationarity. By contrast, the deterministic Fisher/Cramér–Rao bounds in Sec. III additionally assume approximately Gaussian processed noise in the analysis band; when strong non-Gaussian tails or obvious nonstationarity remain, the safer quantity to quote is the spectrum-level bound in Eq. (34) rather than the ideal matched-filter error bar.

At microscopic level, however, many candidate models are first written in terms of a force-density or momentum-diffusion kernel rather than a torque directly. For a torsional mode about $\hat{z}$ one has schematically

$$\begin{aligned} \tau_z(t) &= \int d^3r [\mathbf{r} \times \mathbf{f}(\mathbf{r}, t)]_z, \\ S_{\tau\tau}^{\text{sc}}(\omega) &= \int d^3r d^3r' \epsilon_{zij} r_i \epsilon_{zkl} r'_k S_{f_j f_l}^{\text{sc}}(\mathbf{r}, \mathbf{r}'; \omega). \end{aligned} \quad (38)$$

with repeated Cartesian indices summed, or the corresponding discrete-mass sum. A force-diffusion model therefore maps to D_τ only after the geometry/form-factor integral implied by Eq. (38) is carried out; in a rigid-body lever-arm estimate this reduces schematically to $D_\tau \sim \mathcal{G}_{\text{geom}} l_{\text{eff}}^2 D_F$, where $\mathcal{G}_{\text{geom}} = \mathcal{O}(1)$ and D_F is the white diffusion coefficient of the generalized force projected onto the relevant translational mode, defined by $S_{FF}(\omega) = 4D_F$ and therefore carrying units $\text{N}^2 \text{s}$.

This intermediary role appears in Oppenheim-type classical-quantum gravity and in its weak-field elaborations. There the fundamental diffusion is formulated for classical gravitational variables, and the induced torque spectrum of a specific torsion mode depends on geometry, coarse-graining, and readout conventions rather than on a universal setup-independent number [17, 18]. There is therefore no single postquantum “target” value of D_τ that can be quoted independently of the apparatus; producing one requires exactly the projection in Eq. (38) for

the suspension and source geometry at hand. Our D_τ is the experiment-facing quantity that any concrete theory must map onto before comparison with data, and a full Oppenheim-to-torsion mapping for realistic pendula is left to future work. The same theory-to-experiment workflow already appears in the collapse-model literature, where rotational-noise data have been translated into geometry-dependent CSL exclusions for both tabletop and space-based platforms [24, 25].

C. Equivalent torque spectra as the natural comparison space

For the stochastic problem we divide the measured angle spectrum by $|\chi|^2$ and work in equivalent torque units,

$$S_{\tau\tau}^{\text{eq}}(\omega) \equiv \frac{S_{yy}(\omega)}{|\chi(\omega)|^2}. \quad (39)$$

Equation (39) collapses any torque-like source into a directly comparable quantity. In practice, one should divide by the magnitude of the calibrated response actually used in the subtraction; for a structural-loss calibration this means $|\chi| \rightarrow |\chi_\phi|$. White viscous thermal noise becomes a flat line [Eq. (13)], structural damping becomes the $1/\omega$ spectrum in Eq. (16), and a white semiclassical diffusion term is then a constant $4D_\tau$. Accordingly, Fig. 2 below is arguably more informative for semiclassical-gravity constraints than the raw angle ASD.

V. QUANTITATIVE BENCHMARKS

A. Representative room-temperature Cavendish benchmark

The numbers in this subsection are an illustrative scale-setting model, not a universal prediction for all torsion balances. They are chosen to make the conversion between mechanical parameters, angle spectra, equivalent torque spectra, SQL benchmarks, and white torque-diffusion levels transparent.

We begin with a benchmark chosen to display the scalings explicitly:

$$\begin{aligned} I &= 5.0 \times 10^{-5} \text{ kg m}^2, & \kappa &= 1.0 \times 10^{-7} \text{ N m/rad}, \\ Q &= 5.0 \times 10^3, & T &= 300 \text{ K}. \end{aligned} \quad (40)$$

The corresponding resonance is

$$\begin{aligned} f_0 &= \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{\kappa}{I}} \simeq 7.12 \text{ mHz}, \\ \theta_{\text{zpf}} &\simeq 4.86 \times 10^{-15} \text{ rad}. \end{aligned} \quad (41)$$

For the simple two-end-mass picture one may regard this as $I = 2ml^2$ with $m = 10 \text{ g}$ and $l = 5 \text{ cm}$, but in the rest of the paper I is always treated as the measured

mechanical parameter rather than inferred from a point-mass model.

The resonant susceptibility is

$$|\chi(\omega_0)| = \frac{Q}{I\omega_0^2} = 5.0 \times 10^{10} \text{ rad/(N m)}. \quad (42)$$

To define a reference quantum-limited detector, we choose white imprecision and white backaction spectra saturating Eq. (10) and touching the SQL at resonance:

$$S_{\theta\theta}^{\text{imp}} = \hbar |\chi(\omega_0)|, \quad S_{\tau\tau}^{\text{ba}} = \frac{\hbar}{|\chi(\omega_0)|}. \quad (43)$$

Numerically,

$$\begin{aligned} \sqrt{S_{\theta\theta}^{\text{imp}}} &\simeq 2.30 \times 10^{-12} \text{ rad}/\sqrt{\text{Hz}}, \\ \sqrt{S_{\tau\tau}^{\text{ba}}} &\simeq 4.59 \times 10^{-23} \text{ N m}/\sqrt{\text{Hz}}. \end{aligned} \quad (44)$$

The measurement-added SQL at resonance is therefore

$$\sqrt{S_{\theta\theta}^{\text{SQL}}(\omega_0)} = \sqrt{2\hbar |\chi(\omega_0)|} \simeq 3.25 \times 10^{-12} \text{ rad}/\sqrt{\text{Hz}}. \quad (45)$$

Equivalently, $S_{\theta\theta}^{\text{SQL}}(\omega_0) = 4\theta_{\text{zpf}}^2 Q/\omega_0$, which leaves θ_{zpf} as a single-oscillator benchmark even though the main text works directly with calibrated spectra.

By contrast, the one-sided viscous thermal torque PSD is

$$S_{\tau\tau}^{\text{th,visc}} = 4k_B T I \gamma = 7.41 \times 10^{-30} \text{ N}^2 \text{ m}^2/\text{Hz}, \quad (46)$$

corresponding to a resonant angle ASD

$$\sqrt{S_{\theta\theta}^{\text{th,visc}}(\omega_0)} = 1.36 \times 10^{-4} \text{ rad}/\sqrt{\text{Hz}}. \quad (47)$$

Thus the benchmark thermal floor exceeds the measurement-added SQL by a factor $1.36 \times 10^{-4}/(3.25 \times 10^{-12}) \simeq 4.2 \times 10^7$ in ASD, i.e. about 7.6 orders of magnitude (or $\sim 1.8 \times 10^{15}$ in PSD). This large gap is precisely why current torsion-balance experiments are not usually limited by quantum backaction, even though the SQL remains the correct asymptotic benchmark once classical torque noise is reduced.

Figure 1 shows the linear calibrated angle-ASD budget, including both viscous and structural thermal models. The corresponding viscous thermal diffusion scale is

$$D_\tau^{\text{bench}} = \frac{S_{\tau\tau}^{\text{th,visc}}}{4} = 1.85 \times 10^{-30} \text{ N}^2 \text{ m}^2 \text{ s}. \quad (48)$$

Equation (48) is not an experimental semiclassical-gravity bound; it is the illustrative stochastic torque level that a residual analysis would need to beat on this representative room-temperature platform.

An experimentalist should read Fig. 1 as an angle-domain diagnostic. A narrowband deterministic torque template at frequency ω_m is constrained by taking the local ordinate $\sqrt{S_{yy}(\omega_m)}$ from the relevant processed spectrum and inserting it in Eq. (28). The figure also shows

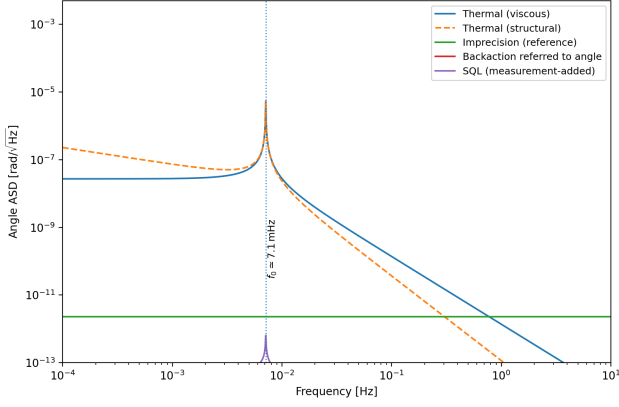


FIG. 1. Benchmark angle ASDs for the representative Cavendish parameters in Eq. (40). The thermal curves are shown for both viscous and structural damping. The imprecision and backaction curves are the reference pair of Eq. (43), tuned to touch the measurement-added SQL at resonance. The values at $f_0 \simeq 7.1$ mHz are numerically consistent with Eqs. (44)–(47).

why reducing the optical imprecision alone does not automatically improve a millihertz gravitational test: if the observed local floor is thermal or environmental, the matched-filter gain is controlled by that classical floor until it is subtracted or physically reduced.

B. A public stochastic benchmark from the 2025 torsion-balance search

A more directly experimental benchmark is available from the recent torsion-balance search of Yan *et al.*, who reported an ultralow-frequency optomechanical platform with $f_0 = 0.6$ mHz, quality factor exceeding 5×10^4 , and an angle sensitivity of $0.3 \mu\text{rad}/\sqrt{\text{Hz}}$ at 2.5 mHz [8]. Using the reported torsional moment of inertia $I = 0.14 \text{ kg m}^2$ for the same rotational mode [8, 28], the susceptibility magnitude at 2.5 mHz is

$$|\chi(2.5 \text{ mHz})| \simeq 3.07 \times 10^4 \text{ rad}/(\text{N m}). \quad (49)$$

Treating the published $0.3 \mu\text{rad}/\sqrt{\text{Hz}}$ as a conservative bound on the *total* observed floor, the corresponding equivalent torque ASD is

$$\sqrt{S_{\tau\tau}^{\text{eq}}} = \frac{0.3 \times 10^{-6} \text{ rad}/\sqrt{\text{Hz}}}{|\chi(2.5 \text{ mHz})|} \simeq 9.77 \times 10^{-12} \text{ N m}/\sqrt{\text{Hz}}, \quad (50)$$

which implies the conservative observed-floor white-diffusion bound

$$D_{\tau}^{\text{Yan}} \lesssim \frac{S_{yy}^{\text{obs}}}{4|\chi|^2} \simeq 2.38 \times 10^{-23} \text{ N}^2 \text{ m}^2 \text{ s}. \quad (51)$$

Equation (51) is conservative because it uses a quoted observed sensitivity floor rather than a fully subtracted

residual semiclassical-noise spectrum. We use this single public sensitivity point as a reference worked example; a full reanalysis of the published spectrum would employ Eq. (37) over the full residual band. Using only the minimum published quality factor, the quoted resonance frequency, and a room-temperature estimate gives a reference viscous thermal scale for the same platform,

$$D_{\tau,\text{th}}^{\text{Yan},\text{min}} \equiv k_B T I \frac{\omega_0}{Q} \simeq 4.4 \times 10^{-29} \text{ N}^2 \text{ m}^2 \text{ s}, \quad (52)$$

where we used $Q = 5 \times 10^4$, $f_0 = 0.6$ mHz, and $T = 300$ K. This is about 5×10^5 times smaller than Eq. (51), consistent with Yan *et al.*'s statement that thermal noise was not the dominant observed floor near 2.5 mHz [8]. The public conservative bound therefore does not reduce to the platform's thermal floor rewritten in different units.

As a theory-facing orientation point, Eq. (51) may also be read as a bound on a white generalized-force diffusion,

$$D_F \lesssim \frac{D_{\tau}^{\text{Yan}}}{l_{\text{eff}}^2} \simeq 2.4 \times 10^{-19} \left(\frac{l_{\text{eff}}}{1 \text{ cm}} \right)^{-2} \text{ N}^2 \text{ s}, \quad (53)$$

for an effective lever arm l_{eff} . Device-specific theory comparisons must then supply the appropriate form factor of Eq. (38). Rotational-noise analyses have already been mapped into specific collapse-model parameter bounds. In particular, Altamura *et al.* showed that a tabletop rotational-noise analysis can constrain CSL near $\lambda \sim 10^{-9} \text{ s}^{-1}$ at $r_C = 10^{-4} \text{ m}$ and that geometry optimization can substantially tighten the bound, while rotational noise from LISA Pathfinder strengthens the constraints further [24, 25]. We do not translate those results into a universal D_{τ} , because the mapping depends on the detailed mass-density form factor of the experimental geometry, but they illustrate exactly why the intermediary experimental quantity should be a calibrated torque spectrum rather than a theory-specific parameter. The same logic underlies direct semiclassical-source proposals such as optomechanical tests of the field generated by a quantum superposition and Schrödinger–Newton-induced dephasing analyses: the theory is first posed in native source parameters, while the experiment responds to a geometry-dependent force or torque observable that must be projected onto the measured mode [22, 23].

Figure 2 places the representative room-temperature benchmark curves and the Yan-specific reference on common axes. The benchmark curves set the scale of measurement-added and thermal torque noise for Eq. (40), while the public Yan point and the illustrative viscous thermal reference of Eq. (52) are overlaid in the same equivalent-torque units. The overlay is intended as a unit-consistent comparison between benchmark and platform-specific quantities, not as a statement that both datasets arise from a single suspension model. At sufficiently low frequency, the equivalent torque ASD of a structurally damped fiber rises as $1/\sqrt{f}$ and can be misread as an exotic stochastic drive if the suspension model is not independently characterized.

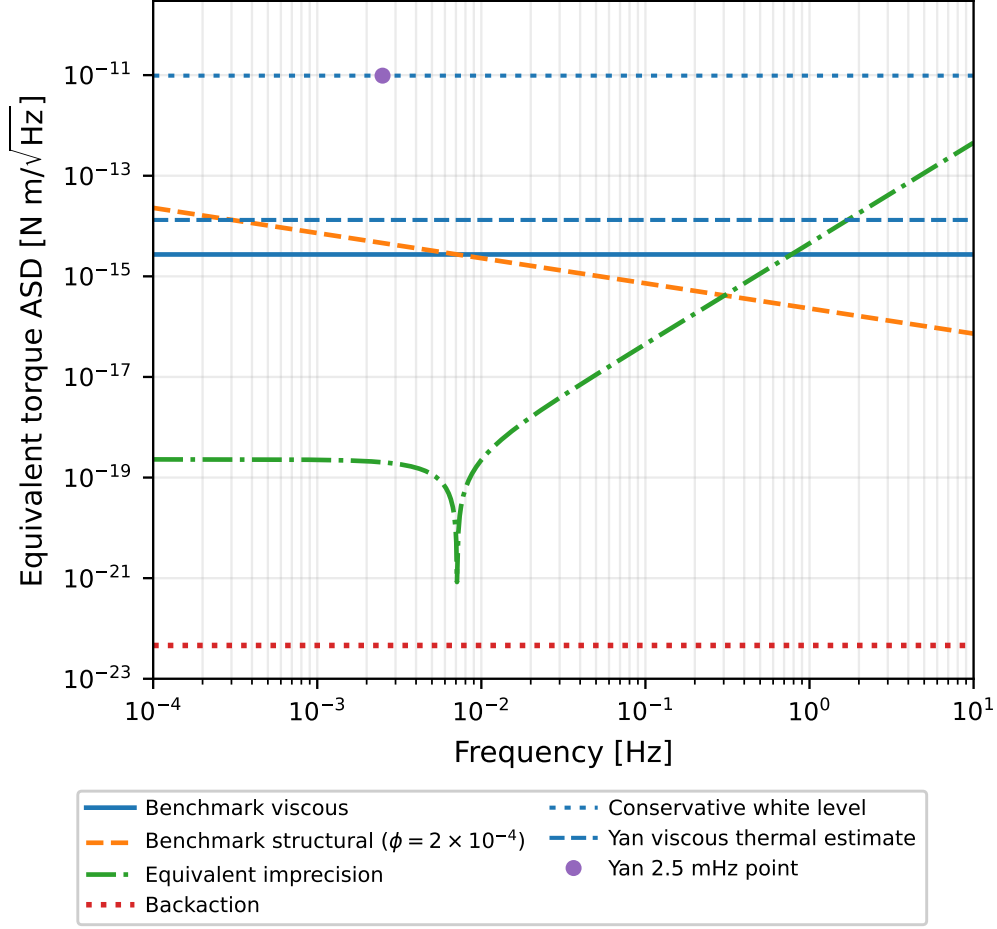


FIG. 2. Equivalent torque ASDs on common axes. The benchmark curves show, in equivalent torque units, the representative room-temperature viscous thermal, structural thermal, equivalent-imprecision, and backaction levels for Eq. (40); the structural-loss curve uses $\phi = 2 \times 10^{-4}$ only as an illustration of the red $1/\sqrt{f}$ scaling in equivalent torque units. The point marks the public sensitivity quoted by Yan *et al.* at 2.5 mHz, converted by Eq. (50). The dashed horizontal line shows the illustrative viscous thermal level inferred from the published f_0 , Q , and I via Eq. (52); the dotted horizontal line shows the conservative white level obtained by interpreting the public point as a frequency-independent floor. The common-axis overlay places benchmark and platform-specific quantities in the same equivalent-torque units; it does not model a single common suspension.

For experimental design, Fig. 2 is the more direct plot for stochastic models. A candidate model predicting a torque spectrum $S_{\tau\tau}^{\text{model}}(\omega)$ should be plotted against the residual equivalent-torque spectrum rather than against the raw angle ASD. A flat model can be summarized by D_τ , but a colored model is constrained only where its full frequency dependence lies below the residual upper band. The figure also shows that an observed public floor and a thermal estimate for the same platform answer different questions: the former is an experimental observed-floor bound, while the latter is a suspension-noise scale inferred from a simplified damping model.

C. Classical-source benchmarks from Eöt-Wash

The strongest existing torsion-balance tests of the WEP constrain composition-dependent differential accelerations at the part-in- 10^{13} level [4], while inverse-square-law searches exclude gravitational-strength Yukawa interactions down to tens of micrometers [5, 6]. We do not attempt to re-extract those limits from raw spectra here. Instead, the point relevant for semiclassical gravity is simpler: any candidate classical-source modification must reproduce the same null results when projected onto the corresponding torque templates $\Lambda_x(t)$. In other words, the deterministic matched-filter framework already built into standard torsion-balance null tests is exactly the correct language for the deterministic sector of semiclassical-gravity phenomenology.

VI. PAGE–GEILKER AND MEP AS MATCHED-FILTER APPLICATIONS

This section is an interpretive application of the general template formalism, not a claim that Page–Geilker branch discrimination and Fedida–Kent-style MEP tests are physically equivalent. They differ in state-preparation logic and in the physical sourcing rule being tested. The common structure appears only after each competing prediction has been projected onto the torque that the torsion balance would see. For any two candidate sourcing rules A and B , let

$$\Delta\tau(t) \equiv \tau_A(t) - \tau_B(t). \quad (54)$$

The optimal Gaussian discrimination statistic is governed by

$$\text{SNR}_{A/B}^2 = 4 \int_0^\infty \frac{d\omega}{2\pi} \frac{|\chi(\omega)\Delta\tau(\omega)|^2}{S_{yy}(\omega)}. \quad (55)$$

The narrowband formula below is the single-line version of Eq. (55). Thus Page–Geilker and MEP tests reduce to the same criterion only after their different physical sourcing assumptions have been converted into the torque difference $\Delta\tau$.

A. What Page–Geilker excludes

Consider two macroscopically distinct source configurations $|L\rangle$ and $|R\rangle$ that produce torques $\pm\tau_0(t)$ on the balance. A naive Møller–Rosenfeld mean-field theory sources gravity from the expectation value of the mass distribution. For the symmetric superposition $(|L\rangle + |R\rangle)/\sqrt{2}$ this implies an approximately vanishing single-run torque, whereas branch-correlated behavior would produce responses $\pm\tau_0$. The Page–Geilker result is therefore often interpreted as excluding the simplest deterministic, no-collapse mean-field sourcing rule on individual runs [7, 16]. It does *not* by itself exclude every semiclassical model: collapse-assisted or stochastic versions can evade the contradiction by producing definite branchwise source configurations or additional diffusion [16, 17].

B. A quantitative branch-discrimination criterion

If one compares a branch-correlated torque $\pm\tau_0 \cos \omega_m t$ to an averaged or vanishing mean-field response at the same modulation frequency, the relevant signal-to-noise ratio is just the narrowband matched-filter expression

$$\text{SNR}_{\text{branch}} \simeq \frac{|\chi(\omega_m)| \tau_0}{\sqrt{S_{yy}(\omega_m) B_w}}. \quad (56)$$

If the sign or phase of the branch-correlated response is not fixed a priori and one performs a two-sided test over

both quadratures, the corresponding detection threshold acquires the same $\sqrt{2}$ factor discussed below Eq. (28). For the representative thermal benchmark of Sec. V, choosing $\omega_m = \omega_0$, a rectangular window, and $T_{\text{int}} = 1$ day yields

$$\tau_{0,\text{min}} \sim \frac{\sqrt{S_{yy}(\omega_0)/T_{\text{int}}}}{|\chi(\omega_0)|} \approx 9 \times 10^{-18} \text{ N m} \quad (57)$$

for unit signal-to-noise if the observed floor is thermal. The numerical value follows by using $|\chi(\omega_0)| = Q/\kappa = 5.0 \times 10^{10} \text{ rad/(N m)}$ and $\sqrt{S_{yy}(\omega_0)} = 1.36 \times 10^{-4} \text{ rad}/\sqrt{\text{Hz}}$ from Eq. (47), so that the coherent one-day angle uncertainty is $1.36 \times 10^{-4}/\sqrt{86400} \simeq 4.6 \times 10^{-7} \text{ rad}$. Equation (57) should not be overinterpreted: it is a benchmark calculation showing how a projected branch template would be tested, not a proposal for an immediately feasible macroscopic superposition experiment or a geometry-specific Page–Geilker/MEP design.

C. Proper and improper mixtures

Fedida and Kent recently emphasized that Møller–Rosenfeld semiclassical gravity violates a weak form of the mixture equivalence principle: proper and improper mixtures with the same density matrix need not be gravitationally equivalent in such a theory [19]. The torsion-balance formulation converts that statement into a comparison of torque templates.

Define

$$\rho_{\text{proper}} = \frac{1}{2}(|L\rangle\langle L| + |R\rangle\langle R|), \quad (58)$$

$$|\Psi\rangle_{SA} = \frac{1}{\sqrt{2}}(|L\rangle_S|0\rangle_A + |R\rangle_S|1\rangle_A), \quad (59)$$

$$\rho_{\text{improper}} = \text{Tr}_A(|\Psi\rangle\langle\Psi|), \quad (60)$$

so that $\rho_{\text{proper}} = \rho_{\text{improper}}$ as density operators on the source subsystem. If gravity respects mixture equivalence, then an unconditional torsion-balance measurement cannot distinguish the two preparations. Operationally, the distinction is between two classes of sourcing rules. A reduced-state rule assigns an unconditional torque functional $\tau_{\text{red}}(t) = \mathcal{T}[\rho_S](t)$ and therefore necessarily obeys $\tau_{\text{red}}[\rho_{\text{proper}}] = \tau_{\text{red}}[\rho_{\text{improper}}]$. A branchwise rule instead assigns $\tau_b(t) = b\tau_0(t)$ with $b = \pm 1$ on each run and only vanishes after ensemble averaging. Any observable difference between the two preparations therefore presupposes a nonstandard gravitational sourcing rule that is *not* a function solely of the reduced density operator ρ_S ; ordinary linear quantum mechanics coupled only through ρ_S would predict no unconditional distinction. In one representative MEP-violating class of mean-field models discussed by Fedida and Kent, the improper preparation is taken to source the averaged mass distribution and hence yields a near-zero unconditional torque, whereas the proper mixture yields a run-by-run

bimodal response $\pm\tau_0$. The discrimination criterion is then again Eq. (56); the only new element is the state-preparation requirement.

Two remarks follow. First, the original Page–Geilker protocol is related to but not identical with a full MEP test, because it does not compare proper and improper mixtures with the same reduced density matrix. Second, the bottleneck in a future MEP experiment is not the statistical analysis, which follows the same matched-filter criterion, but the source preparation: one must realize a source whose gravitational torque is large enough to satisfy Eq. (56) while maintaining control over uncontrolled decoherence and classical randomness. The torsion-balance formalism developed here isolates that requirement.

VII. EXPERIMENTAL DESIGN IMPLICATIONS

The unified deterministic–stochastic framework yields four experimental design implications.

First, the public data product directly used by the bounds in Secs. III and IV is a calibrated residual *equivalent torque* spectrum rather than only a raw angle ASD. In the absence of strong imprecision–backaction correlations, the residual can be written schematically as

$$S_{\tau\tau}^{\text{eq, res}}(\omega) \simeq \frac{S_{yy}^{\text{obs}}(\omega)}{|\chi(\omega)|^2} - S_{\tau\tau}^{\text{th}}(\omega) - \sum_j |K_j(\omega)|^2 S_{x_j x_j}(\omega) - \frac{S_{\theta\theta}^{\text{imp}}(\omega)}{|\chi(\omega)|^2} - S_{\tau\tau}^{\text{ba}}(\omega). \quad (61)$$

where $K_j \equiv \partial\tau/\partial x_j$ converts each auxiliary environmental channel into an equivalent torque contribution. Equation (61) is the diagonal version. If the auxiliary channels are correlated—as they often are when several monitors observe the same Newtonian-noise field—the subtraction should instead use the full cross-spectral matrix,

$$S_{\tau\tau}^{\text{aux}}(\omega) = \sum_{i,j} K_i(\omega) K_j^*(\omega) S_{x_i x_j}(\omega). \quad (62)$$

Thus the separation of noises is a calibrated multichannel inference problem, not an algebraic identity. Low-frequency excess backgrounds — seismic tilt, thermal drift, gravity-gradient fluctuations driven, for example, by atmospheric pressure fields or slow local mass redistribution (including hydrological changes), magnetic pickup, and other $1/f$ -like technical noise — enter through these auxiliary channels or, if left unsubtracted, remain part of the residual itself [15, 27]. When the candidate stochastic signal is itself colored toward low frequency, independent barometric, gravimetric, tilt, or hydrological monitors provide independent constraints, because atmospheric and local-mass Newtonian-noise channels can otherwise be spectrally degenerate with the target signal. If this degeneracy cannot be broken, the correct claim is a conservative bound based on the

observed total floor rather than a separated residual. Ref. [15] gives a complementary analysis of atmospheric Newtonian-noise propagation for torsion-balance G measurements. Such a spectrum can be inserted immediately into Eqs. (36) and (37).

The single formula covers all auxiliary channels. For example, x_j may be platform tilt, barometric pressure, a gravity-gradient monitor, a hydrological proxy, magnetic field, charge or electric-field monitor, temperature, or a deliberately injected calibration line. What matters is not the label of the channel but the measured cross-spectral matrix and the torque calibration $K_j(\omega)$ used to map that channel into equivalent torque units.

Second, any published residual should be accompanied by the metadata needed to reproduce it: independent identification of I , κ , and Q from ringdown and driven transfer-function measurements; the window, ENBW, and leakage treatment entering the spectrum estimate; and the auxiliary calibrations K_j and cross spectra used to subtract the dominant environmental channels. Appendix E summarizes this recipe compactly.

Third, structural damping must be characterized independently. If one extrapolates a room-temperature torsion balance with the wrong loss model into the millihertz band, the resulting equivalent torque floor can be wrong by orders of magnitude. A minimal characterization program is to measure the quality factor as a function of amplitude and temperature, then compare the inferred $S_{\tau\tau}^{\text{eq}}$ with the white prediction of Eq. (13) and the $1/\omega$ prediction of Eq. (16).

Fourth, quantum-limited readout remains relevant even if present experiments are dominated by classical torque noise. The reason is not that the SQL itself is the target of the current semiclassical-gravity bounds, but that any future attempt to sharpen deterministic branch discrimination or stochastic diffusion constraints will eventually require a readout whose imprecision, back-action, and classical environmental couplings have been separately identified. Appendix D records a generic two-channel variational or speed-meter-like strategy. In the present paper it serves as an extension of the classical-noise model, not as a replacement for it.

VIII. CONCLUSIONS

This paper gives a weak-field operational map from calibrated torsion-balance spectra to constraints on deterministic torque templates and additional stationary stochastic torque noise. The result is not a direct test of the full relativistic semiclassical Einstein equation. It is a Newtonian-limit measurement interface: a candidate sourcing rule is projected to a torque waveform or torque-noise spectrum, and that object is compared with the calibrated angular data.

The formal framework consists of two reusable equations. For deterministic effects, any torque template $\tau_x(t) = g_x \Lambda_x(t)$ is bounded by the matched-filter Fisher

information in Eq. (24), with the finite-time, window-aware form in Eq. (28). For stochastic effects, any additional stationary torque noise is bounded by the residual equivalent-torque spectrum in Eq. (34). The scalar coefficient D_τ is only the Markovian white-noise compression of this spectrum; colored or non-Markovian models must be compared with the full frequency-dependent residual upper band. Page–Geilker branch tests and Fedida–Kent mixture-equivalence tests are interpretive applications of this framework once their physically different sourcing rules are expressed as torque templates, not a claim of physical equivalence between the two experiments.

The numerical results have three distinct roles. The representative room-temperature Cavendish calculation is an illustrative benchmark: it shows that, for the chosen parameters, the resonant thermal angle ASD exceeds the measurement-added SQL by about 4.2×10^7 in amplitude and that the corresponding viscous thermal white-diffusion scale is $D_\tau^{\text{bench}} \simeq 1.9 \times 10^{-30} \text{ N}^2 \text{ m}^2 \text{ s}$. This benchmark is useful for scale-setting but is not an experimental semiclassical-gravity limit. The Yan *et al.* conversion is different: the published $0.3 \mu\text{rad}/\sqrt{\text{Hz}}$ point at 2.5 mHz gives a conservative observed-floor white-diffusion bound of order $10^{-23} \text{ N}^2 \text{ m}^2 \text{ s}$, not a fully subtracted residual semiclassical-noise constraint. The inferred Yan thermal scale near $10^{-29} \text{ N}^2 \text{ m}^2 \text{ s}$ is a suspension-noise orientation point based on the published f_0 , Q , and I , and should not be conflated with the observed-floor bound.

Figures 1 and 2 show how these numbers should be used by experimentalists. The angle-ASD plot identifies the local floor that enters the finite-time matched-filter uncertainty for a deterministic template. The equivalent-torque plot is the more natural comparison space for stochastic gravitational models: a white model can be summarized by a horizontal $4D_\tau$ level, whereas a colored model must be compared frequency by frequency with the residual equivalent-torque spectrum. In both cases the design target is an operational band, chosen where the calibrated Fisher kernel is large and where Newtonian, thermal, technical, and quantum readout noises can be separated.

Application of these expressions requires calibrated residual spectra, suspension-loss characterization, and auxiliary-channel transfer functions, including cross-spectral information when several monitors observe the same Newtonian-noise field. With those ingredients, torsion-balance data can be reported in a form that supports direct reuse in deterministic template tests and stochastic semiclassical-gravity searches. Deterministic Newtonian-noise sensing, source classification, recovery-time analysis, and inverse reconstruction are natural extensions of the same formalism, but they require geometry-specific templates and auxiliary channels and are best developed as a separate follow-up study.

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Appendix A: One-sided spectral conventions and the structural-loss FDT

Because the manuscript combines quantum measurement noise, thermal noise, and stochastic semiclassical drive noise, the spectral convention must be fixed once and for all. For a real stationary process $x(t)$ we define the one-sided PSD $S_{xx}(\omega)$ for $\omega > 0$ by

$$\langle x(\omega)x^*(\omega') \rangle = 2\pi \delta(\omega - \omega') S_{xx}(\omega), \quad \omega, \omega' > 0, \quad (\text{A1})$$

and the ASD is $\sqrt{S_{xx}}$.

With the Fourier convention used in the main text and the structural-loss response $\chi_\phi(\omega) = [\kappa(1 - i\phi) - I\omega^2]^{-1}$, the one-sided fluctuation–dissipation relation for the angle coordinate is

$$S_{\theta\theta}^{\text{th, str}}(\omega) = \frac{4k_B T}{\omega} \text{Im} \chi_\phi(\omega). \quad (\text{A2})$$

Since

$$\text{Im} \chi_\phi(\omega) = \frac{\kappa\phi}{(\kappa - I\omega^2)^2 + (\kappa\phi)^2}, \quad (\text{A3})$$

one immediately obtains Eq. (16) after dividing by $|\chi_\phi|^2$. The corresponding double-sided symmetrized expressions differ by the familiar factor of two.

Appendix B: White torque diffusion and angular-momentum variance

This appendix fixes the stochastic-process meaning of D_τ . Let the stochastic torque in the white Markovian limit be denoted by $\xi_\tau(t)$ and write the angular-momentum equation as

$$\dot{L}(t) = \xi_\tau(t), \quad \langle \xi_\tau(t)\xi_\tau(t') \rangle = 2D_\tau \delta(t - t'), \quad (\text{B1})$$

where the correlation function is understood as the double-sided time-domain convention. Then

$$\begin{aligned} \langle [L(t) - L(0)]^2 \rangle &= \int_0^t dt_1 \int_0^t dt_2 \langle \xi_\tau(t_1)\xi_\tau(t_2) \rangle \\ &= 2D_\tau t. \end{aligned} \quad (\text{B2})$$

Thus D_τ is the coefficient controlling random-walk growth of angular-momentum variance. The double-sided white PSD corresponding to Eq. (B1) is $2D_\tau$, and the one-sided PSD used in the main text is therefore

$$S_{\tau\tau}^{1s}(\omega) = 4D_\tau, \quad \omega > 0. \quad (\text{B3})$$

If the torque noise is colored, the angular-momentum variance is instead determined by the full correlation function or full spectrum; no single D_τ captures the process.

Appendix C: Finite-time matched filtering

For completeness we sketch the finite-time normalization leading to Eq. (28) in the single-tone case. Consider data

$$y(t) = A \cos(\omega_m t + \varphi) + n(t), \quad 0 < t < T_{\text{int}}, \quad (\text{C1})$$

where $n(t)$ is stationary Gaussian noise with one-sided PSD $S_{yy}(\omega_m)$ that is approximately constant across the bandwidth of the analysis window. For a chosen window $w(t)$, define the normalized quadrature estimator

$$\hat{A} \equiv \frac{2}{\int_0^{T_{\text{int}}} dt w(t)} \int_0^{T_{\text{int}}} dt w(t) y(t) \cos(\omega_m t + \varphi). \quad (\text{C2})$$

Averaging $\cos^2$ over many cycles gives $\langle \hat{A} \rangle \simeq A$. For the same many-cycle approximation, the variance of the estimator is

$$\begin{aligned} \text{Var}(\hat{A}) &\simeq \frac{4}{\left[\int_0^{T_{\text{int}}} dt w(t) \right]^2} \left(\frac{S_{yy}(\omega_m)}{2} \right) \\ &\quad \times \int_0^{T_{\text{int}}} dt w^2(t) \cos^2(\omega_m t + \varphi), \end{aligned} \quad (\text{C3})$$

and hence

$$\text{Var}(\hat{A}) \simeq S_{yy}(\omega_m) \frac{\int_0^{T_{\text{int}}} dt w^2(t)}{\left[\int_0^{T_{\text{int}}} dt w(t) \right]^2} = S_{yy}(\omega_m) B_w. \quad (\text{C4})$$

With the one-sided convention used in the main text, the factor $1/2$ from the many-cycle average $\langle \cos^2 \rangle = 1/2$ is exactly cancelled by the factor relating the local real-quadrature noise variance to the one-sided PSD. Equivalently, the same known-phase result may be written as $\delta A = \sqrt{2S_{yy}^{\text{sym}}(\omega_m) B_w}$ in double-sided symmetrized notation. Therefore

$$\delta A \simeq \sqrt{S_{yy}(\omega_m) B_w}. \quad (\text{C5})$$

For a rectangular window one has

$$\delta A = \sqrt{\frac{S_{yy}}{T_{\text{int}}}}, \quad (\text{C6})$$

whereas for a Hann window

$$\delta A = \sqrt{\frac{3S_{yy}}{2T_{\text{int}}}}. \quad (\text{C7})$$

If the phase is unknown and must be estimated from two orthogonal quadratures, the variance doubles and δA acquires the usual factor $\sqrt{2}$. If the signal frequency is not exactly tracked, a leakage factor $\mathcal{L} \leq 1$ should be inserted as $A \rightarrow \mathcal{L}A$.

Appendix D: Generic two-channel variational readout

The main text does not require any specific readout architecture beyond Eq. (7), but we record the generic form of a two-channel estimator because future semiclassical-gravity experiments may need such an upgrade. The phrase “variational readout” means that the experiment does not use the raw angle channel alone. It also measures an auxiliary quadrature or auxiliary channel that carries information about the noise that will otherwise appear as backaction. A frequency-dependent linear combination can then cancel part of the correlated noise without canceling the signal.

Suppose two angle-equivalent channels are available,

$$y_1(\omega) = \theta(\omega) + n_1(\omega), \quad (\text{D1})$$

$$y_2(\omega) = c_s(\omega)\theta(\omega) + c_b(\omega)\tau_{\text{ba}}(\omega) + n_2(\omega), \quad (\text{D2})$$

where n_1 and n_2 are the intrinsic noises of the two readout channels. The coefficient $c_s(\omega)$ is the calibrated signal transfer from the true angle to the auxiliary channel, and $c_b(\omega)$ is the calibrated transfer from the back-action torque to that channel; their units are whatever is required to make y_2 angle equivalent. The second channel could be realized, for example, by a speed-meter-like optical response, an auxiliary cavity monitoring radiation-pressure correlations, or a stroboscopic quadrature measurement. The cross spectra below are defined by $S_{ij}(\omega) \equiv S_{N_i N_j}(\omega)$ for the noise parts N_i of the two channels after the known signal response has been removed. Consider the estimator

$$\theta_{\text{est}}(\omega) = y_1(\omega) + \alpha(\omega)y_2(\omega). \quad (\text{D3})$$

Its noise PSD is

$$S_{\text{estest}}(\omega) = S_{11}(\omega) + |\alpha(\omega)|^2 S_{22}(\omega) + 2\text{Re}[\alpha(\omega)S_{12}(\omega)]. \quad (\text{D4})$$

Minimizing over α gives the closed-form weight

$$\alpha_{\text{opt}}(\omega) = -\frac{S_{12}(\omega)}{S_{22}(\omega)}, \quad (\text{D5})$$

and the corresponding minimum

$$S_{\text{estest}}^{\text{min}}(\omega) = S_{11}(\omega) - \frac{|S_{12}(\omega)|^2}{S_{22}(\omega)}. \quad (\text{D6})$$

This is the torsion-balance counterpart of standard variational readout in interferometric optomechanics. The crucial point is that the auxiliary channel must be *physically specified* and calibrated through c_s , c_b , and the cross spectrum S_{12} ; a formal reference to the canonical momentum operator alone is not sufficient for an experimental proposal. In a concrete optical realization c_s and c_b are generally not independent free functions: both are linked by the same cavity or transducer dynamics, and the role of the generic treatment here is only to state

the optimal linear estimator once those coupled transfer functions have been calibrated. In practice, S_{12} should be measured with the same windowing and leakage treatment used in the science analysis, ideally using coherent calibration lines and optical-power sweeps that isolate the backaction pathway from purely electronic correlations.

Appendix E: Minimal publication recipe for reusable residual spectra

For future torsion-balance papers intended to support semiclassical-gravity reinterpretation, the minimum data product required for reuse is a residual equivalent-torque spectrum together with the metadata needed to reconstruct it. This appendix is a reporting checklist, not an additional physical assumption: it states what has to be published so that another group can tell which part of an observed spectrum is thermal, Newtonian, readout, or still-unexplained residual noise. The checklist is:

1. identify I , κ , and Q from ringdown and/or driven transfer-function measurements;
2. state the PSD convention (one-sided or double-sided), the window, the ENBW, and any leakage correction $\mathcal{L}$ used in the estimator;

3. provide the suspension-loss calibration, including the frequency band over which a constant loss angle ϕ is a valid approximation;
4. publish the dominant auxiliary channels x_j and their torque calibrations $K_j = \partial\tau/\partial x_j$ so that Eq. (61) can be reconstructed;
5. when a fully subtracted residual is unavailable, state explicitly whether the quoted bound is based on a residual spectrum or on the observed total floor.

If a white-noise summary D_τ is quoted, it is also appropriate to state the leading calibration uncertainty. A simple first-order estimate is

$$\frac{\delta D_\tau}{D_\tau} \simeq \left[\left(2 \left\langle \frac{\delta|\chi|}{|\chi|} \right\rangle_{\mathcal{B}} \right)^2 + \left(\left\langle \frac{\delta S_{yy}^{\text{res},+}}{S_{yy}^{\text{res},+}} \right\rangle_{\mathcal{B}} \right)^2 \right]^{1/2}, \quad (\text{E1})$$

where the band average follows the same weighting used in Eq. (37). This estimate assumes that the susceptibility calibration and the residual-spectrum estimate are effectively independent; if the same suspension-model parameters enter both, their covariance should be propagated explicitly rather than neglected. This checklist records the minimum metadata needed to make the quantities in Secs. III and IV portable across platforms.

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